

IT06 Final Project

G5

Original: an origami game

1



Fold a sheet of paper

2



Create a paper airplane

3

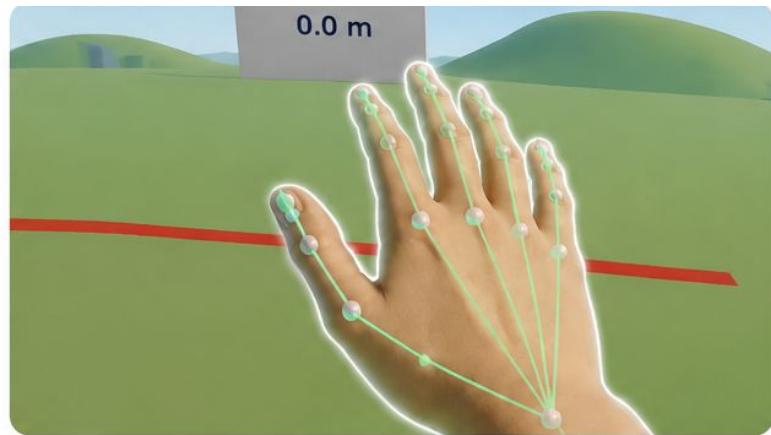
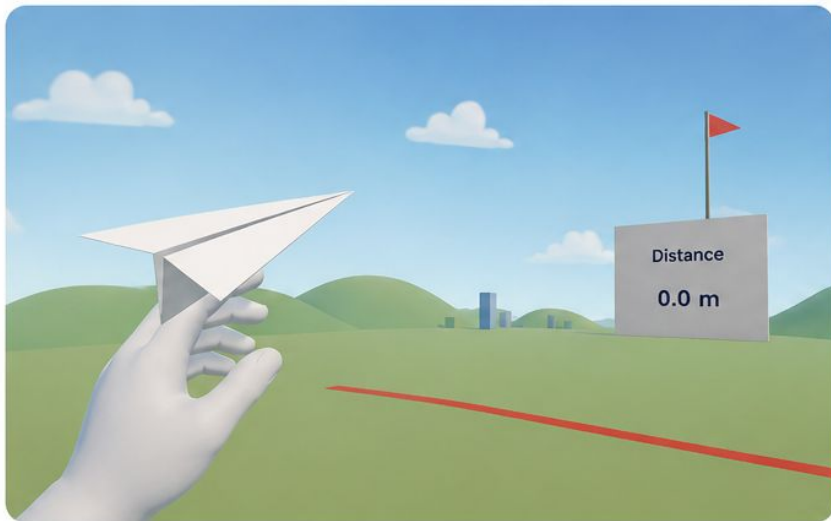


Throw it

4



Measure distance



Problem

Paper folding simulation was too difficult.



Inspiration & New Direction

In ex12, we learned XR Hands and hand-tracking interaction.



We decided to create a game that fully utilizes hand-tracking interaction!

Game Concept: Stick Catch Game

1



START

Press the start button to begin the game.



2



Random Stick Falls

A random stick falls from the top.



3



Catch Stick

Catch the falling stick using both hands.



4



Get Score

You get +1 point for each successful catch.



5



FINAL SCORE

When all sticks are processed, the final score is displayed.



Goal

Catch as many falling sticks as possible before they hit the floor!

Main Features

Hand Tracking

- XR Hands
- OpenXR



Grabbing

- XR Grab Interactable



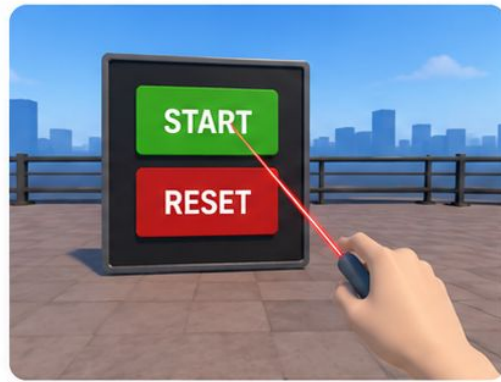
Physics

- Rigidbody
- Gravity
- Collision Detection



Raycasting

- Laser Pointer UI



Additional Systems & Original Content



Score System

- Real-time score update
- Final score display



Audio Feedback

- Catch sound
- Button sound
- Final score sound



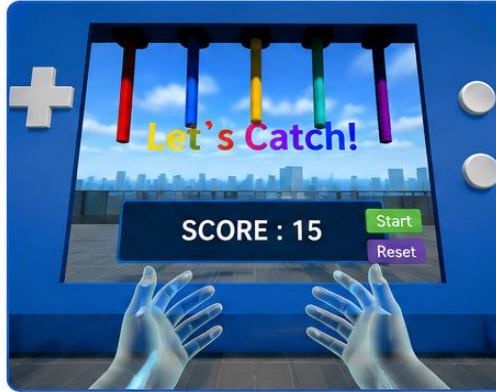
Visual Feedback

- Glow effect
- Machine color change
- Shake animation



Original Content

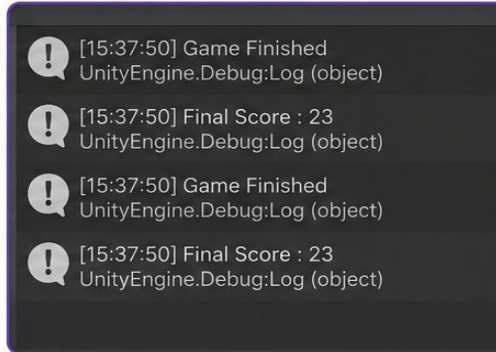
- Custom game machine
- Buildings and tower



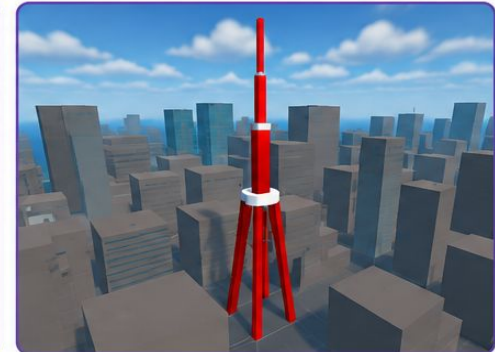
Score System (In-Game UI)



Visual Feedback (Glow & Color Change)



Debug Script (Console Log)



Original Content (Environment)

Technologies & Scripts

Technologies



Unity XR



Hands



XR Interaction
Toolkit

etc...

Main Scripts



Stick.cs

Controls stick behavior
and detection



StickManager.cs

Manages stick spawning
and timing



GameManager.cs

Handles score and
game flow



GameMachine.cs

Controls visual feedback
effects

```
> Images
> Objects
> Samples
> Scenes
> Scripts
  C# ButtonSE.cs
    ButtonSE.cs.meta
  C# GameManager.cs
    GameManager.cs.m...
  C# GameManager.cs
    GameManager.cs.m...
  C# Stick.cs
    Stick.cs.meta
  C# StickManager.cs
    StickManager.cs.meta
> SE
> Settings
> TextMesh Pro
> TutorialInfo
> Video
> XR

4 using UnityEngine.XR.Interaction.Toolkit;
5
6 public class Stick : MonoBehaviour
7 {
8     private Rigidbody rb;
9     private XRGrabInteractable grab;
10
11     private Renderer stickRenderer;
12     private Material stickMaterial;
13
14     private AudioSource audioSource;
15
16     private Vector3 startPosition;
17     private Quaternion startRotation;
18
19     private bool isCaught = false;
20     private bool isDropping = false;
21
22     public bool hasDropped = false;
23
24     void Start()
25     {
26         rb = GetComponent<Rigidbody>();
27
28         grab = GetComponent<XRGrabInteractable>()
```



Summary

These technologies and scripts work together to create an interactive and immersive stick catch game.

Live Demo & Conclusion

What We Learned

- ✓ XR Hands
- ✓ Unity Physics
- ✓ VR Interaction Design
- ✓ Game System Integration

Why VR is Fun

- ✓ Immersive Experience
- ✓ Natural Hand Interaction
- ✓ Real-Time Feedback

LET'S DEMO!